

Practice session

Department of Mathematics

Name: _____

1 Exponential equations**1.1 Examples**

1 To solve the equation $5^{2x} = 5^{x+1}$ we can use the fact that is a bijective function, that is, is one to one. Since is a bijective function and have the same base, we can equal the exponents,

$$2x = x + 1.$$

Now, we can apply algebra tools to solve for x ,

$$\begin{aligned} (2x) - x &= (x + 1) - x \\ \cancel{2x} - \cancel{x}^x &= \cancel{x}^x - \cancel{x}^0 + 1 \\ x &= 1. \end{aligned}$$

Hence, the answer is $x = 1$. To fact check the answer, we can substitute the value in the original equation,

$$\begin{aligned} 5^{2x} \Big|_{x=1} &= 5^{x+1} \Big|_{x=1} \\ 5^{2(1)} &= 5^{1+1} \\ 5^2 &= 5^2. \end{aligned}$$

Since both side of the equal sign are the same value we check that $x = 1$ is a solution of the equation.

This example is a special case, so now we are going to explore an other example to see if we can get a general and robust methodology to solve this type of equations.

2 Consider the following equation $3^{x+2} = 7$. For this case we are going to use the Product law of logarithms $\log(x^a) = a \log(x)$. First, we apply the log function in both sides of the equation,

$$\begin{aligned} 3^{x+2} &= 7 \\ \log(3^{x+2}) &= \log(7) \\ (x+2) \log(3) &= \log(7) \\ (x+2) &= \frac{\log(7)}{\log(3)} \\ x+2 &= \frac{\log(7)}{\log(3)} \\ x &= \frac{\log(7)}{\log(3)} - 2. \end{aligned}$$

Check the answer by computing the numeric value using your calculator.

Guidelines for solving Exponential equations From the last example we can abstract the procedure to the following steps to solve any exponential equation,

- Isolate the exponential expression on one side of the equation.
- Take the logarithm of each side, then use the Laws of logarithms to reduce the expression.
- Use algebra tools to solve for the variable.

1.1.1 Some notes

Example 1 We can repeat the procedure of the second example for the first example as follows,

$$\begin{aligned} 5^{2x} &= 5^{x+1} \\ \log(5^{2x}) &= \log(5^{x+1}) \\ (2x) \log(5) &= (x+1) \log(5) \\ \frac{(2x)}{(x+1)} &= \frac{\log(5)}{\log(5)} \quad \begin{matrix} \nearrow 1 \\ \searrow 1 \end{matrix} \\ \frac{(2x)}{(x+1)} &= 1 \\ 2x &= 1(x+1) \\ 2x - x &= 1 \\ x(2-1) &= 1 \\ x &= 1. \end{aligned}$$

As we can see, we got the same result as before. Even though the procedure seems to be more laborious, is much more robust than the one used in the first example.

Example 2 In this example we saw how we can use the logarithms properties to solve an exponential equation. However we use the logarithmic function with base 10 ($\log$). Now we are going to explore to use the fact of inverse function between the logarithmic and exponential functions. Recalling that $\log_a(a^x) = x$ and that $a^{\log_a(x)} = x$, we can solve the second example as follows,

$$\begin{aligned} 3^{x+2} &= 7 \\ \log_3(3^{x+2}) &= \log_3(7) \\ (x+2)\log_3(3) &= \log_3(7), \end{aligned}$$

analysing the factor $\log_3(3)$, we can use the property of the inverse function, $\log_3(3) = 1$,

$$\begin{aligned} (x+2)\log_3(3) &\stackrel{1}{=} \log_3(7) \\ x+2 &= \log_3(7) \\ x &= \log_3(7) - 2. \end{aligned}$$

If we cant to compute the numeric value of that expression we can use the change of base formula,

$$\begin{aligned} x &= \log_3(7) - 2 \\ &= \frac{\log(7)}{\log(3)} - 2. \end{aligned}$$

Which is exactly the same value as before.

1.2 Exercises

Solve the following equations,

1. $8e^{2x} = 20$
2. $e^{3-2x} = 4$
3. $e^{2x} - e^x - 6 = 0$
4. $3xe^x + x^2e^x = 0$

Be careful with exercises 2 and 4.

2 Logarithmic equations

2.1 Examples

1 To solve the following equation

$$\log(x^2 + 1) = \log(x - 2) + \log(x + 3)$$

we are going to reduce the right hand side of the equation using properties of logarithms,

$$\log(x^2 + 1) = \log(x - 2) + \log(x + 3)$$

$$\log(x^2 + 1) = \log[(x - 2)(x + 3)].$$

Now, we are going to use the fact that the logarithm is a bijective function. Therefore, to solve that equation, the arguments needs to be equal (as the example 1 of exponential functions). Hence, we get the following polinomial equation,

$$\begin{aligned} x^2 + 1 &= (x - 2)(x + 3) \\ x^2 + 1 &= x^2 - 2x + 3x - 6 \\ x^2 - x^2 - x &= -6 - 1 \\ -x &= -7 \\ x &= 7. \end{aligned}$$

If we evaluate the equation at $x = 7$ we can reassure that is the solution of the equation.

2 Now, we are going to solve the following equation $\log_2(25 - x) = 3$. For this equation we can not use the same argument of example 1, because there is no logarithm function in both sides. Hence we are going to apply the property of inverse functions, $a^{\log_a(x)} = x$ and $\log_a(a^x) = x$.

First we are going to apply the following change of variable $\alpha = 25 - x$, allowing us to re-write the equation as follows,

$$\log_2(25 - x) = 3 \rightarrow \log_2(\alpha) = 3.$$

Analyzing the equation we know that is a logartihm base 2, so in order to use the inverse function property we need to apply an exponential function base 2 in both sides,

$$\begin{aligned} 2^{\log_2(\alpha)} &= 2^3 \\ \alpha &= 2^3, \end{aligned}$$

now, we undo the change of variable and apply algebra tools to solve for x ,

$$\begin{aligned} 25 - x &= 2^3 \\ -x &= 2^3 - 25 \\ x &= \cancel{25} - \cancel{2^3}^{17} \\ &= 17. \end{aligned}$$

Looking at that procedure we can get a general procedure:

Guidelines for solving logarithmic equations

- Isolate the logarithmic term on one side of the equation. (Reduce the logarithms)
- Write the equation in exponential form.
- Use algebra to solve for the variable.

2.1.1 Some notes

Example 1 We are going to use the same procedure from example 2 into example 1. First we reduce the expression using laws of logarithms,

$$\begin{aligned} \log(x^2 + 1) &= \log(x - 2) + \log(x + 3) \\ \log(x^2 + 1) &= \log((x - 2)(x + 3)) \end{aligned}$$

Since both sides cannot be reduce an further, we apply an exponential function base 10 in both sides,

$$10^{\log(x^2+1)} = 10^{\log((x-2)(x+3))},$$

then, using the properties of the inverse functions $a^{\log_a(x)} = x$,

$$\begin{aligned} x^2 + 1 &= (x - 2)(x + 3) \\ x^2 + 1 &= x^2 - \cancel{2x} + \cancel{3x}^x - 6 \\ \cancel{x^2} - \cancel{x^2}^0 - x &= \cancel{-6} - \cancel{1}^{-7} \\ -x &= -7 \\ x &= 7. \end{aligned}$$

Which is the same result the we previously computed.

2.2 Exercises

Solve the following equations,

1. $4 + 3 \log(2x) = 16$

2. $\log(x+2) + \log(x-1) = 1$

3. $\ln(x) = 10$

4. $\log_2(x) + \log_2(x-3) = 1$